Probabilități

(1)

- Examen ← Proiect – scris de mână
  întrebări
- Bibliografie –
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- Definiția axiomatică
- Prob. condiționate
  - Formula probabilității totale
  - Formula Bayes

- Scheme clasice de probab
  - Schema Bernoulli (simplă)
  - Schema Binomială (cazul bilei revenite)
  - Schema Multinomială (.... revenite)
  - Schema Poisson
  - Schema hipergeometrică (cazul bilei nerevenite)

- Variabile aleatoare (asociate)
  - cazul discret
  - cazul absolut continuu
- Caracteristici asociate v.a.
  - valoare medie
  - dispersie
  - momente de diferite ordine
- Legi de probab ← discrete / continue ...
- LNM – legea numerelor mari
- TLC – teorema limită centrală
- funcție caracteristică, fc. generatoare

$0.833 = \sqrt{\frac{n_1 n_2}{n_1+n_2}} D_{n_1,n_2} < 1.91 \Rightarrow$ cele două selecții provin din aceeași populație (acceptăm H_0)

- zar $\begin{cases} 1 - 4 - 6 \\ \frac{A}{6} \end{cases}$ (2)

fată cu nr par: 2 4 6 $\quad \frac{3}{6} = \frac{1}{2}$

- moneda $\Omega = \{S, V\}$ evenimente elementare

- 2 monede $\Omega = \{S\,S, \begin{matrix} S\,V \\ V\,S \end{matrix}, V\,V\}$ $\quad \frac{2}{4} = \frac{1}{2}$.

- 2 zaruri $\Omega = \{(1,1), (1,2) \ldots (1,6), (2,1) \ldots (6,1) \ldots, (6,6)\}$

Ω eveniment sigur.

Def. axiomatică

Ω, (E) $\quad$ (K) $\mathcal{P}(\Omega) = \{\phi, \ldots, \Omega\}$

moneda $\Omega = \{S, V\}$ $\quad \mathcal{P}(\Omega) = \{\phi, \{S\}, \{V\}, \underbrace{\{S, V\}}_{\Omega}\}$

Repartiția beta

$f(x) = \begin{cases} \frac{x^{p-1}(1-x)^{q-1}}{B(p,q)}, & x \in [0,1] \\ 0, & x \notin [0,1] \end{cases}$, unde $B(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}$

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Zar: $S(\Omega) = \{\phi, \{1\}, \dots \{6\}, \{1,2\}, \dots \{1,6\}, \dots \{1,2,\dots,6\}\}$.

$\underline{P} : \mathcal{K} \longrightarrow [0,1]$

$\mathcal{K}$

- $\Omega, \phi \in \mathcal{K}$.
- $A \in \mathcal{K} \Rightarrow CA \in \mathcal{K}$ ($\bar{A}$) ($\mathcal{K}$ este închisă la complementară)
- $A_1, A_2, \dots \in \mathcal{K} \Rightarrow A_1 \cup \dots \cup \dots \in \mathcal{K}$.
  I cel mult numărabilă
  $\bigcup_{i \in I} A_i \in \mathcal{K}$.

$A, B \in \mathcal{K}$.
$A \cup B \in \mathcal{K} \Rightarrow C(A \cup B) \in \mathcal{K}$
$\| \quad CA \cap CB$.
$\in \mathcal{K}$

Def

- $P(\Omega) = 1$
- $0 \leq P(A) \leq 1 \quad \forall A \in \mathcal{K}$.
- $A, B \in \mathcal{K} \Rightarrow P(A \cup B) = P(A) + P(B)$
  $A \cap B = \phi$ ____ incompatibile.

Zar $A = \{1,2\}$, $B = \{4,5,6\}$ ≤ 2

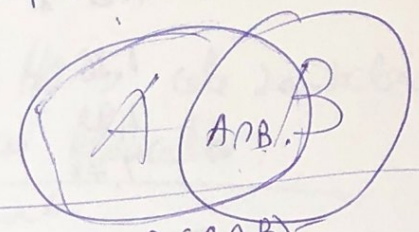
4.

A, B, C.

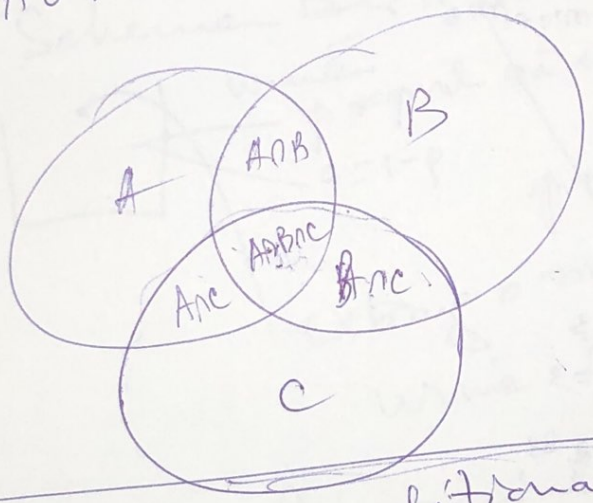
$A \cap B = \emptyset$
$A \cap C = \emptyset$
$B \cap C = \emptyset$
$\Rightarrow P(A \cup B \cup C) =$
$= P(A) + P(B) + P(C)$

A, B

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$



$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B)$
$- P(A \cap C) - P(B \cap C)$
$+ P(A \cap B \cap C)$.



Prob. condiționată

A, B.

$A = (A \cap B) \cup (A \cap \overline{B})$

(A)

$A = A \cap \Omega = A \cap (B \cup \overline{B}) =$
$= (A \cap B) \cup (A \cap \overline{B})$

B $\overline{B}$

$2 \cdot (3+5) \neq 2 \cdot 5$
$2 \cdot 3$
$(2 \cdot 3)(4 \cdot 5)$

$A_1 = A \cap B$, $A_2 = A \cap \overline{B}$

$A_1 \cap A_2 = A \cap B \cap A \cap \overline{B} =$
$= A \cap B \cap \overline{B} = \emptyset$.

$n_1=3 \quad n_2=2 \quad n_3=3$

$\Rightarrow t = \frac{S_1}{2} : \frac{S_{12}}{5} = 5,93$

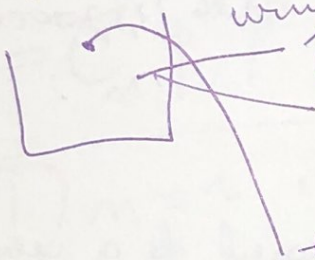
(5)

$\alpha = 0,01 \quad \Rightarrow F_{0,99}(2,5) = 13,27$.

Deoarece $t < F_{0,99}(2,5)$ nu se respinge $H_0 \Rightarrow$ cele 3 antibiotice nu fel eficiente asupra bacteriei Klebsiella.

Scheme de probe

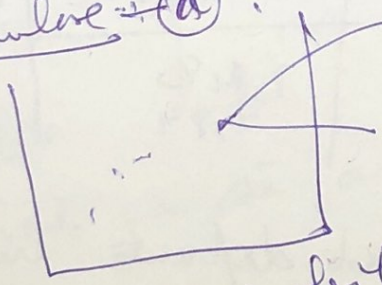
1) Schema Bernoulli



urnă
p = prob să extrag o bilă albă
$q = 1-p$.
$p+q=1 \quad p,q \geq 0$.
extrag o singură bilă din urnă și GATA!

2) Schema Binomială (schemă bilei întoarsă)

o formulă ⇒ (a).



p = "albă"
q = "neg"
experiență.

Se extrag n bile din urnă (una de fiecare dată bila extrasă la loc (extrageri independente)
Probl: Care este prob să în cele n extrageri să avem k bile albe?

A A - - A N A A N N - - -

p p p next

k albi

p . . . p q . . . q

k n-k

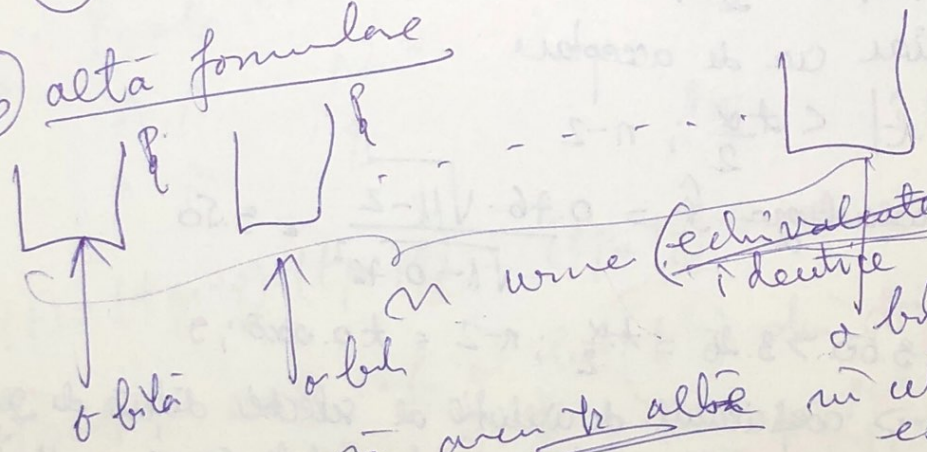
N A A N

N - - - A

$$p_n(k) = C_n^k p^k q^{n-k}$$

(!) $n=1 \Rightarrow$ urna Bernoulli-ştiută.

(b) altă formulare



Pb?: Prob ca să avem k albe în cele n extrase!

$C_n^k p^k q^{n-k}$?

$$(p+q)^n = \sum_{k=0}^{n} C_n^k p^k q^{n-k} = 1.$$

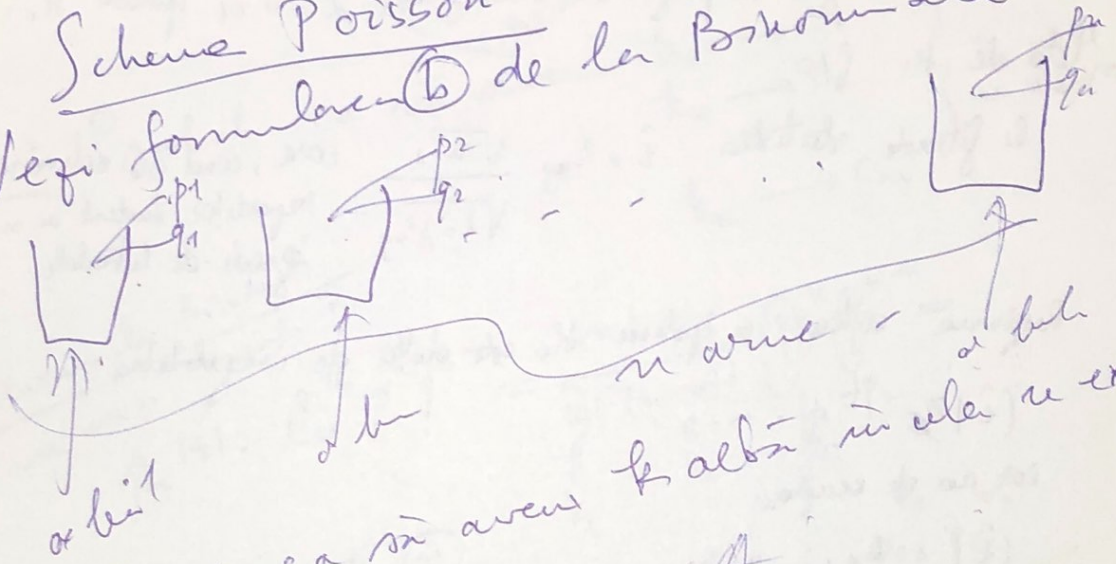
$$\sum_{n=0}^{n} p_n(k) = 1$$

$$f(t) = (tp+q)^n = \sum_{k=0}^{n} t^k \underbrace{C_n^k p^k q^{n-k}}_{p_n(k)}$$

coeficientul lui t^k din dezv. binomului $(t \cdot p+q)^n$

Schema Poisson

Vezi formularea (b) de la Binomială



Pb: Prob ca să avem k albe în cele n extrag

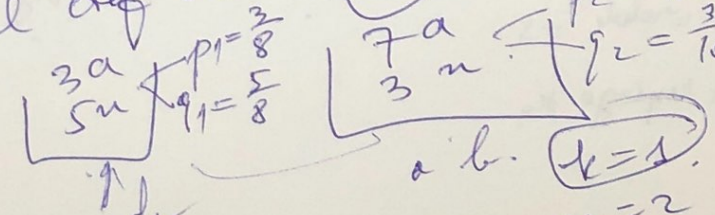
A A ... N - - - A

p_1 p_2 q_5 p_{25}

$f(t) = (tp_1+q_1)\cdot(tp_2+q_2)\cdots(tp_n+q_n) =$ ← aceasta

Prob = acel coef al lui t^k din dez.

Ex:

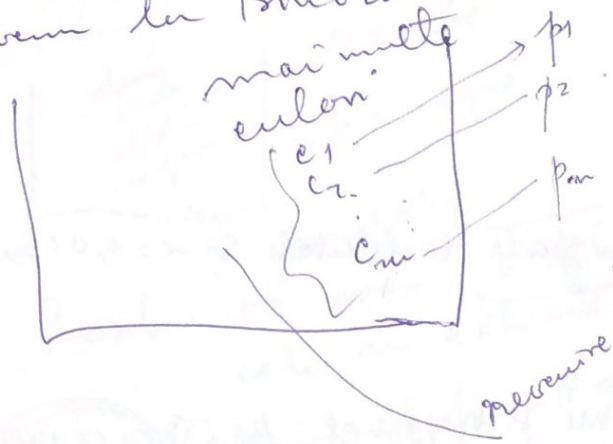


$$f(t) = \left(\frac{3}{8}t+\frac{5}{8}\right)\left(\frac{7}{10}t+\frac{3}{10}\right) = \cdots t\left(\frac{3}{8}\cdot\frac{3}{10}+\frac{5}{8}\cdot\frac{7}{10}\right)$$

Prob cerută!

Schema multinomială

Revenim la Bernoulli: mai multe evenimente



$p_1 + \dots + p_m = 1$.

extragem n bile pe rând de fiecare dată punând la loc bila extrasă

Pb: Probab ca să avem k_1 apariții ale lui c_1, k_2 apariții ale lui c_2, ..., k_m apariții ale lui c_m } în cele n extrase

Revenim la binomială:
Cum era prob?

$$p_n(k) = C_n^k p^k q^{n-k} = \frac{n!}{k!\,(n-k)!} p^k q^{n-k} =$$

($k = k_1$, $n-k = k_2$, $p = p_1$, $q = p_2$)

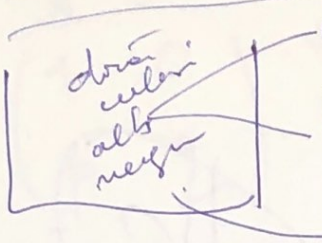
$k_1 + k_2 = k + n - k = n$

$k_1 + k_2 + \dots + k_m = n$.

$$= \frac{n!}{k_1!\,k_2!} p_1^{k_1} p_2^{k_2}.$$

$$\boxed{p_n(k_1, k_2, \dots, k_m) = \frac{n!}{k_1!\,k_2! \cdots k_m!} p_1^{k_1} p_2^{k_2} \cdots p_m^{k_m}.}$$

Schema hipergeometrică
Schema Bernoulli - cu bila nereveniță

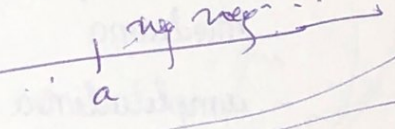


extragem n bile

Pb: Prob. să avem k albe, $n-k$ negre. în cele n extrase

Aici folosim definiția clasică a probabilității

$$p_n(k) = \frac{\text{nr. caz. favorabile}}{\text{nr. caz posibile}} = \frac{C_a^k \cdot C_b^{n-k}}{C_{a+b}^n}$$



$0 \le k \le a$
$0 \le n-k \le b$
$n \le a+b$

Variabile aleatoare

$X : \begin{pmatrix} S & V \\ p & q \end{pmatrix}$ v.a.

$X : \Omega \to \mathbb{R}$, (Ω, K) $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$

$X(\omega)$

$\Omega = \{S, V\}$

$X(S) = 1$

$X(V) = 0$

1 extrag

1 1 0 0 0 1 1 1 0 0 0

frecvente

$M(X) = 1 \cdot p + 0 \cdot q$

E

$X : \begin{pmatrix} x_1 & x_2 & \dots & x_n \\ p_1 & p_2 & \dots & p_n \end{pmatrix}$ repartitie a v.a. X

$M(X) = x_1 p_1 + \dots + x_n p_n = \sum_{i=1}^{n} x_i p_i$

$X - M(X)$

$\mu_i \geq 0$, $\sum \mu_i = 1$

$M(X) = m$

v.a. discreta

$D^2(X) = M(X - M(X))^2$

$M(X)$

$\sum_{i=1}^{n} (x_i - m)^2$

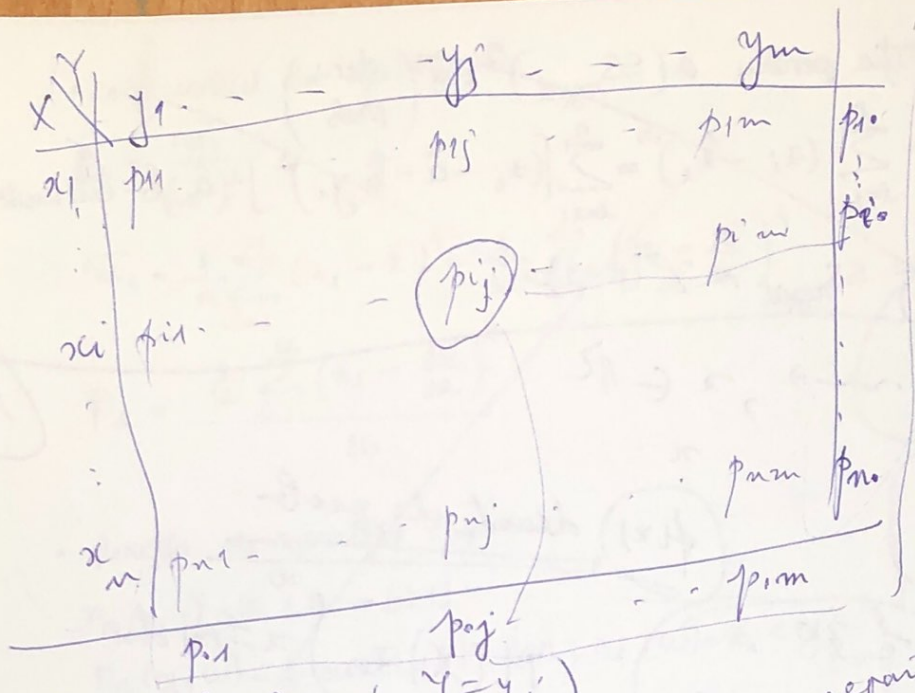
$D(X)$?

Varianta = $Var(X)$

$X \overset{not}{=} \begin{pmatrix} x_1 & \dots & x_n \\ p_1 & \dots & p_n \end{pmatrix}$ $Y \overset{not}{=} \begin{pmatrix} y_1 & \dots & y_m \\ q_1 & \dots & q_m \end{pmatrix}$

$\in K$

$p_1 = P(X = x_1) = P(\{\omega \in \Omega / X(\omega) = x_1\})$



$p_{ij} = P(X = x_i, Y = y_j)$

$X : \begin{pmatrix} x_1 & \cdots & x_i & \cdots & x_n \\ p_{1\cdot} & \cdots & p_{i\cdot} & \cdots & p_{n\cdot} \end{pmatrix}$ repartiția marginală a lui X

$Y : \begin{pmatrix} y_1 & \cdots & y_j & \cdots & y_m \\ p_{\cdot 1} & \cdots & p_{\cdot j} & \cdots & p_{\cdot m} \end{pmatrix}$ repartiția marginală a lui Y.

X și Y sunt independente dacă $\forall i, j$

$p_{ij} = p_{i\cdot} \times p_{\cdot j}$

$$M(\ ) = \sum_{i=1}^{n} \sum_{j=1}^{m} x_i p_{ij} y_j$$

c) Repartiția pentru $2(SS_{rezid})^5 + 3\begin{pmatrix} dens \\ prob \end{pmatrix}$

$SS_{rezid} = \sum_{i=1}^{n} (x_i - \hat{x}_i)^2 = \sum_{i=1}^{n} [(x_i - \hat{a} - \hat{b} y_i)^2]$, $\hat{a}, \hat{b}$ - estimatori

$\frac{1}{\sigma^2(1-\rho^2)} SS_{rezid} \sim \chi^2(n-2)$

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$X = x \rightsquigarrow x \in \mathbb{R}^2$.

x

x_i | $f(x)$ densit de prob.
p_i

$f \geq 0$

$\int_{-\infty}^{\infty} f(x)\,dx = 1$

$M(X) = \int_{-\infty}^{\infty} x f(x)\,dx$

$D^2(X) = \int_{-\infty}^{\infty} (x - \underbrace{M(X)}_{m_1})^2 f(x)\,dx$

X v. a. discretă:

$X : \begin{pmatrix} x_1 & \dots & x_i & \dots & x_n \\ p_1 & \dots & p_i & \dots & p_n \end{pmatrix}$.

$X^3 : \begin{pmatrix} x_1^3 & \dots & x_i^3 & \dots & x_n^3 \\ p_1 & \dots & p_i & \dots & p_n \end{pmatrix}$

? X^3

$7X^2 : \begin{pmatrix} 7x_1^2 & \dots & 7x_i^2 & \dots & 7x_n^2 \\ p_1 & \dots & p_i & \dots & p_n \end{pmatrix}$

$X^3 + 7X^2 : \begin{pmatrix} x_1^3 + 7x_1^2 & \dots & x_i^3 + 7x_i^2 & \dots & \sim \\ p_1 & \dots & p_i & \dots & p_n \end{pmatrix}$

• coeficientul de aplatizare:

$$\beta_2 = \frac{\overline{\nu_4}}{(s^2)^2}$$

$$\overline{\nu_4} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^4 = \frac{1}{12}\sum_{i=1}^{12}\left(x_i - \frac{15}{12}\right)^4$$

$$\beta_2 = \frac{\frac{1}{12}\sum_{i=1}^{12}\left(x_i - \frac{15}{12}\right)^4}{16}$$

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• funcția de repartiție empirică:

$F_n(\cdot,\cdot): \mathbb{R} \times \Omega \to [0,1]$

$F_n(y,\omega) = \frac{1}{n}\,\text{card}\{i / 1 \le i \le n,\ x_i(\omega) = x_i < y\}$ (=)

$F_n(y,\omega) = \frac{1}{n}\sum_{i=1}^{n} 1_{\{x_i < y\}}(\omega)$

$M(X^k) = \sum_{i=1}^{\infty} x_i^k p_i$ momentul ordin (k)

Repartiții clasice

I) Repartiții discrete

1) Repartiția uniformă pe o mulțime finită

$X: \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} \end{pmatrix}$ $a_1 < \ldots < a_n$

$X \sim U(a_1, a_2, \ldots, a_n)$

$M(X) = E(X) = a_1 \cdot \frac{1}{n} + a_2 \frac{1}{n} + \ldots + a_n \frac{1}{n} = \frac{1}{n}\sum_{i=1}^{n} a_i$ //

$M(X^2) = E(X^2) = \frac{1}{n}\sum_{i=1}^{n} a_i^2$

$M(X^k) = \frac{1}{n}\sum_{i=1}^{n} a_i^k$

$D^2(X) = \mathrm{Var}(X) = M(X - M(X))^2 =$

$= M\left(X^2 - 2X \cdot M(X) + (M(X))^2\right) =$

$= M(X^2) - 2 \cdot M(X) \cdot M(X) + (M(X))^2 =$

$= M(X^2) - (M(X))^2$

$\boxed{D^2(X) = M(X^2) - (M(X))^2}$

Inct nostm

$D^2(X) = \frac{1}{n}\sum_{i=1}^{n} a_i^2 - \left(\frac{1}{n}\sum a_i\right)^2 =$

$= \frac{n\sum_{i=1}^{n} a_i^2 - (\sum a_i)^2}{n^2}$

Ex $X \sim U\{1, 2, \ldots, 6\}$. zar!

Caz particular: repartitie Dirac $X: \begin{pmatrix} a \\ 1 \end{pmatrix}$

$X \sim \delta_a$

caz. monedei

$\begin{pmatrix} S & V \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ $\begin{pmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

2) Bernoulli: $X: \begin{pmatrix} 1 & 0 \\ p & q \end{pmatrix}$; $X^2: \begin{pmatrix} 1^2 & 0^2 \\ p & q \end{pmatrix}$

$M(X) = p$. $M(X^2) = p$.

$D^2(X) = M(X^2) - (M(X))^2 = p - p^2 = p(1-p) = pq$

$B = A^c_{n,1-\alpha}(\mu_0)$.

• Probabilitatea erorii de tip I: $P_{\mu_0}(B) = \alpha$.

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• Funcția caracteristică operatoare:

$$OC(\mu) = P_\mu(A_{n,1-\alpha}(\mu_0)) = P_\mu\left(-z_{1-\frac{\alpha}{2}} \le \frac{\sqrt{n}(\bar{X}-\mu_0)}{\sigma} \le z_{1-\frac{\alpha}{2}}\right) =$$

16)=

$$= P_\mu\left(-z_{1-\frac{\alpha}{2}} \le \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} + \frac{\sqrt{n}(\mu-\mu_0)}{\sigma} \le z_{1-\frac{\alpha}{2}}\right) =$$

$$= P_\mu\left(-z_{1-\frac{\alpha}{2}} - \frac{\sqrt{n}(\mu-\mu_0)}{\sigma} \le \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \le z_{1-\frac{\alpha}{2}} - \frac{\sqrt{n}(\mu-\mu_0)}{\sigma}\right) =$$

$$= F\left(z_{1-\frac{\alpha}{2}} - \frac{\sqrt{n}(\mu-\mu_0)}{\sigma}\right) - F\left(-z_{1-\frac{\alpha}{2}} - \frac{\sqrt{n}(\mu-\mu_0)}{\sigma}\right).$$

$X \sim Be$

$M(X) = p$.

$D^2(X) = pq$

3) Reg. Bernoulli (bila revenită)

$$p_n(k) = C_n^k p^k q^{n-k}, \quad k \in \{0,1,\dots,n\}$$

v.a X.

$X \sim Bi(n, p)$

prob să extrag bila albă

bila albă care mă interesează

extragem n

$$X : \begin{pmatrix} 0 & 1 & \dots & k & \dots & n \\ q^n & C_n^1 p q^{n-1} & \dots & \underbrace{C_n^k p^k q^{n-k}}_{p_n(k)} & \dots & p^n \end{pmatrix}$$

(17)

$$\begin{cases} p_n(k) \geq 0 \\ \sum_{k=0}^{n} p_n(k) = 1 \end{cases} ? \quad \left(1 = (p+q)^n = \sum_{k=0}^{n} C_n^k p^k q^{n-k} = \sum_{k=0}^{n} p_n(k) \right)$$

Prop Fie $X \sim Bi(n, p)$.

Atunci $m = m_X = np$

$$D^2(X) = \sigma_X^2 = npq$$

dem

a) $m_X = M(X) = E(X) = \sum_{k=0}^{n} k \cdot C_n^k p^k q^{n-k} =$

$$= \sum_{k=1}^{n} k \cdot \frac{n!}{(k-1)! \cdot k \cdot (n-k)!} p^k q^{n-k} = n \cdot \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)! \cdot (n-k)!} p^k q^{(n-1)-(k-1)} =$$

$$= np \sum_{k=1}^{n} C_{n-1}^{k-1} p^{k-1} \cdot q^{(n-1)-(k-1)} =$$

$$= np \cdot \underbrace{(p+q)^{n-1}}_{1} = np$$

b) $D^2(X) = Var(X) = \sigma_X^2 = M(X^2) - (M(X))^2$

$$M(X^2) = \sum_{k=0}^{n} k^2 C_n^k p^k q^{n-k} =$$

$$= \sum_{k=0}^{n} [k(k-1) + k] \cdot C_n^k p^k q^{n-k} =$$

$$\sum_{k=0}^{n} k(k-1)\cdot C_n^k p^k q^{n-k} + \underbrace{\sum_{k=0}^{n} k C_n^k p^k q^{n-k}}_{M(X)} =$$

$$= \sum_{k=2}^{n} k(k-1)\cdot \frac{n(n-1)\cdot(n-2)!}{k(k-1)(k-2)!\,(n-k)!} p^k q^{n-k} + M(X)$$

$$= n(n-1)p^2 \sum_{k\geq 2} \frac{(n-2)!}{(k-2)!\,((n-2)-(k-2))!} p^{k-2} q^{(n-2)-(k-2)} + M(X)$$

$$= n(n-1)p^2 \sum_{k\geq 2} C_{n-2}^{k-2} p^{k-2} q^{(n-2)-(k-2)} + M(X)$$

$$= n(n-1)p^2 \cdot \sum_{k\geq 2} C_{n-2}^{k-2} (p+q)^{n-2} + M(X)$$

$$= n(n-1)p^2 + M(X)$$

$$D^2(X) = n(n-1)p^2 + \underbrace{M(X)}_{np} - (np)^2 =$$

$$= n^2p^2 - np^2 + np - n^2p^2 =$$

$$= np(1-p) = npq$$

Reluam:

$$M(X)=\sum_{k=0}^{n} k\,C_n^k p^k q^{n-k}.$$

$$M(X^2)=\sum_{k=0}^{n} k^2\,C_n^k p^k q^{n-k}.$$

$$g(t)=(tp+q)^n=\sum_{k=0}^{n} t^k C_n^k p^k q^{n-k}$$

- $g(1)=(p+q)^n=1=p_n(k)$
- $g'(t)=n(tp+q)^{n-1}\cdot p=\sum_{k=0}^{n} k\,t^{k-1} C_n^k p^k q^{n-k}$

luăm aici $t=1 \Rightarrow$

$$g'(1)=np=\sum_{k=0}^{n} k\,C_n^k p^k q^{n-k}=M(X)$$

- înmulțim cu t pe $g'(t)$

$$tg'(t)=np\cdot t(tp+q)^{n-1}=\sum_{k=0}^{n} k\,t^k C_n^k p^k q^{n-k}$$

derivăm

$$(tg'(t))'=np\left[(tp+q)^{n-1}+t(n-1)(tp+q)^{n-2}\cdot p\right]=$$

$$=\sum_{k=0}^{n} k^2\,t^{k-1} C_n^k p^k q^{n-k}$$

Luăm $t=1 \Rightarrow$

$$np\left[(p+q)^{n-1}+(n-1)p(p+q)^{n-2}\right]=\sum_{k=0}^{n} k^2 C_n^k p^k q^{n-k}.$$

$$np\left(1+p(n-1)\right)=M(X^2)$$

$$M(X^2)=np(np+1-p)=np(np+q)$$

(20).

Ex Se aruncă un zar de 10 ori. Fie X nr. apariții (6) - Să se calculeze

a1) $P(X=2)$ a5) val medie pt X

a2) $P(X \geq 3)$ a6) dispersia pt X.

a3) $P(3 < X < 5)$

a4) $P(X \leq 8)$

$n = 10$ aruncări

Soluție $X \sim Bi(n,p) = Bi(10, \frac{1}{6})$

$p = \frac{1}{6} \Rightarrow P(X=2) = P(\{\omega \in \Omega / X(\omega) = 2\})$

$= p_n(k) = p_{10}(2) = C_{10}^2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^{10-2} = \dots$

a2) $P(X \geq 3) = 1 - P(X \leq 2) = 1 - (P(X=1) + P(X=2))$ (a1)

$P(X=1) = p_{10}(1) = C_{10}^1 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^{10-1}$

a3) $P(3 < X < 5) = P(X=4) = C_{10}^4 \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^{10-4}$

a4) $P(X \leq 8) = 1 - P(X \geq 9) = 1 - (P(X=9) + P(X=10))$

a5) $X : \begin{pmatrix} 0 & 1 & 2 & 3 & 4 & \dots & 10 \\ & & C_n^k p^k q^{n-k} & & & & \end{pmatrix}$

Știm că

$M(X) = np = 10 \cdot \frac{1}{6} = \frac{10}{6}$

$D^2(X) = npq = 10 \cdot \frac{1}{6} \cdot \frac{5}{6} = \frac{50}{36}$

4) Repartiția geometrică

Fie X nr. de probe Bernoulli până la (și excluzând) prima apariție a succesului

X e v.a. discretă cu valori $\mathbb{N}$

$p_X(k) = q^k p = p(1-p)^k$, k natural

$p_X(k) = P(\underbrace{F \cap F \cap \dots \cap F}_{\text{eșec}} \cap \underbrace{S}_{\text{succes}}) \overset{\text{ind}}{=} P(F)\cdots P(F)\,P(S)$, $P(S) = p$

$\sum_{k=0}^{\infty} p_X(k) = \sum_{k=0}^{\infty} q^k p = p \cdot \sum_{k\geq 0} q^k = p\cdot(1+q+q^2+\dots)$

$= p \cdot \frac{1}{1-q} = \frac{p}{p} = 1$ // $\quad \alpha\, q < 1$

$M(X) = \sum_{k=0}^{\infty} k\, p_X(k) = \sum_{k\geq 1} k \cdot p \cdot q^k =$ ($k=0 \Rightarrow 1$)

$= p \sum k q^k = pq \cdot \sum_{k=1}^{\infty} k\, q^{k-1}$

$g(t) = \sum_{k=0}^{\infty} t^k = 1 + t + t^2 + \dots = \frac{1}{1-t}$, $0 < t < 1$

$g'(t) = \sum_{k=1}^{\infty} k\, t^{k-1} = \frac{f'g - fg'}{g'} = \frac{0 - 1\cdot(-1)}{(1-t)^2} = \frac{1}{(1-t)^2}$

$\sum_{k=1}^{\infty} k\, q^{k-1} \overset{t=q}{=\!=} \frac{1}{(1-q)^2} = \frac{1}{p^2}$

$M(X) = pq \cdot \frac{1}{p^2} = \frac{q}{p}$ //

$$D^2(X) = M(X^2) - (M(X))^2$$

$$M(X^2) = \sum_{k=1}^{\infty} k^2 q^k p = p \sum_{k=1}^{\infty} k^2 q^k$$

$$g(t) = \sum_{k\geq 0} t^k = \frac{1}{1-t}, \quad g'(t) = \frac{1}{(1-t)^2} =$$

$$= \sum_{k\geq 1} k t^{k-1} \quad / \cdot t$$

$$\frac{t}{(1-t)^2} = \sum_{k=1}^{\infty} k t^k \quad \text{derivăm}$$

$$\frac{(1-t)^2 + t \cdot 2(1-t)}{(1-t)^4} = \sum_{k\geq 1} k^2 t^{k-1} \quad / \cdot t$$

$$\frac{(1 - 2t + t^2 + 2t - 2t^2)t}{(1-t)^4} = \sum k^2 t^k$$

$$\frac{(1-t^2)t}{(1-t)^4} = \sum k^2 t^k$$

$$\frac{t(1+t)}{(1-t)^3} = \sum k^2 t^k \qquad t = q$$

$$\sum k^2 q^k = \frac{q(1+q)}{(1-q)^3} = \frac{q(1+q)}{p^3}$$

$$M(X^2) = p \cdot \frac{q(1+q)}{p^3} = \frac{q(1+q)}{p^2}$$

$$D^2(X) = \frac{q(1+q)}{p^2} - \frac{q^2}{p^2} = \frac{q + q^2 - q^2}{p^2} = \frac{q}{p^2}$$

(2) Teorema limită centrală.

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Fie $(X_n)_{n\in\mathbb{N}^*}$ un șir de v.a. independente și la fel repartizate cu $M(X_n)=m$ și $D(X_n)=\sigma^2$ și $Y_n = X_1 + \dots + X_n$

Atunci $\lim_{n\to\infty} F_n(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} dt$

Ex

Se aruncă un zar corect.
Fie X nr. de aruncări până la prima apariție a lui (6). Să se calculeze

a1) $P(X=2)$
a2) $P(X\geq 3)$
a3) $P(3<X\leq 5)$
a4) $P(X<8)$

Probab

- def clasică } prob
- def ax.
- def falsuri / pariuri
- FPT
- F Bayes.

- Scheme probab

- rep. de prob
  - discr.
  - cont

- Ineg. Cebâșev
- Legi ale numerelor mari
- Teoreme limită centrală
- fc. caracteristică / fc-generatoare